

UNIVERSITÀ DEGLI STUDI DI MILANO

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Counterfactual Explanations for Graph Neural Networks in Patient Outcome Prediction

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Patient Similarity Networks (PSNs)

- **Nodes**: patients
- **Edges**: clinical and biomolecular similarities between patients

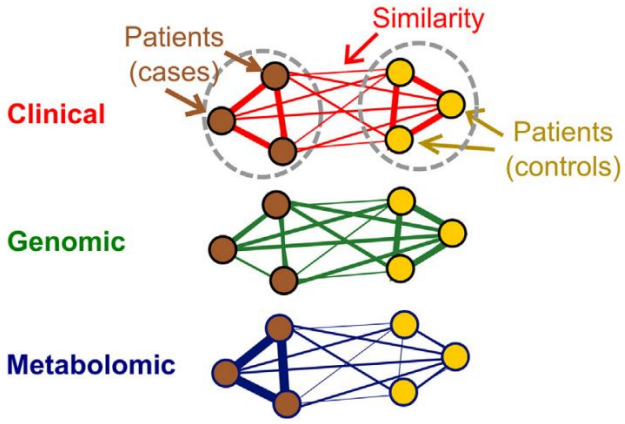


Figure: [Pai '18]

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Overcoming **clinical data challenges:**

- **Multi-modal integration**
  (e.g., via SNF [Wang '14])
- **Low sample size**
  (leveraging graph topology)
- **Missing patient views**
  (recovered via miss-SNF [Gliozzo '25])

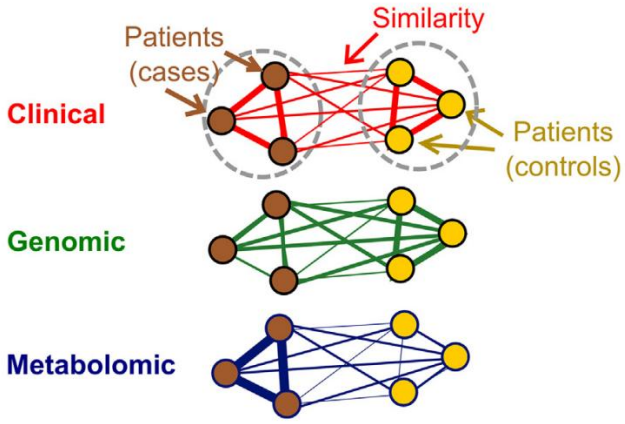


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The Need for XAI in Clinical AI

- **Graph Neural Networks** (GNNs) have **predictive capability**
- GNN **black box** nature **limits** their adoption in the **clinical** setting
- Needs of transparency in decision-making

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Explainable Artificial Intelligence (XAI)

Counterfactual Explanations (CE)

- **Explainable AI** technique, answers *what-if* questions
- Finds the **minimum perturbation** required to alter a model's prediction

"if your BMI were lower than 29, your predicted outcome would be *survived* instead of *deceased*"

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Unlike classical feature selection methods, CE allows for:

- **Instance-level personalization:** Identifying disease drivers specific to the single patient
- **Actionability:** Direct therapeutic intervention

The Gap: Counterfactual Explainers on PSNs

- Tabular CE methods **ignore relational context** (DiCE [Mothilal '20], SingleCF [Wachter '18])
- Graph CE methods focus mostly on **graph classification** (CLEAR [Ma '22])
- Lack of multimodal **integration**

No existing CE algorithm is tailored for *node classification* on PSNs in clinical outcome prediction.

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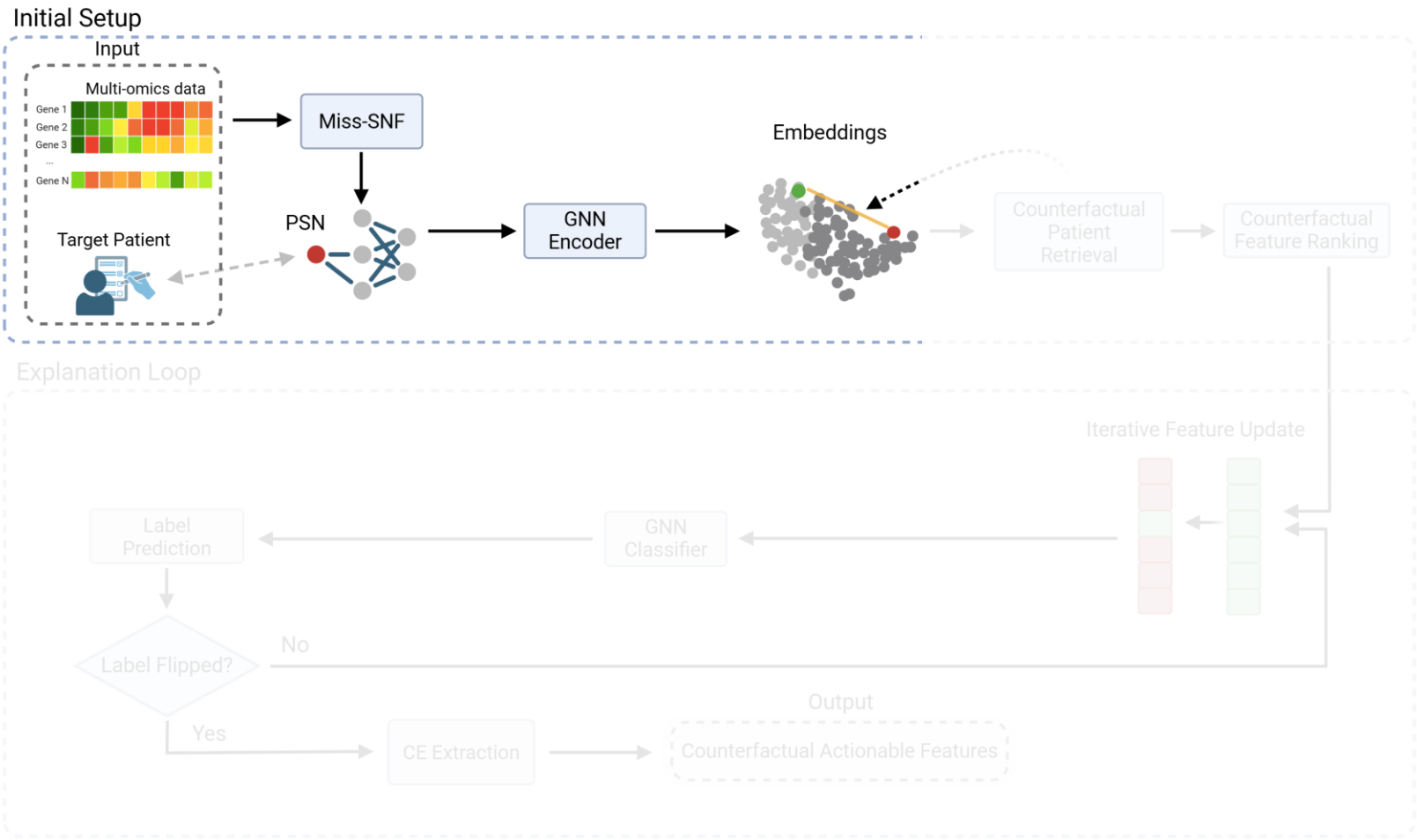
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Our **solution**: **C**ounterfactual **E**xplanations **G**raph **N**eural **N**etworks (**CE-GNN**)

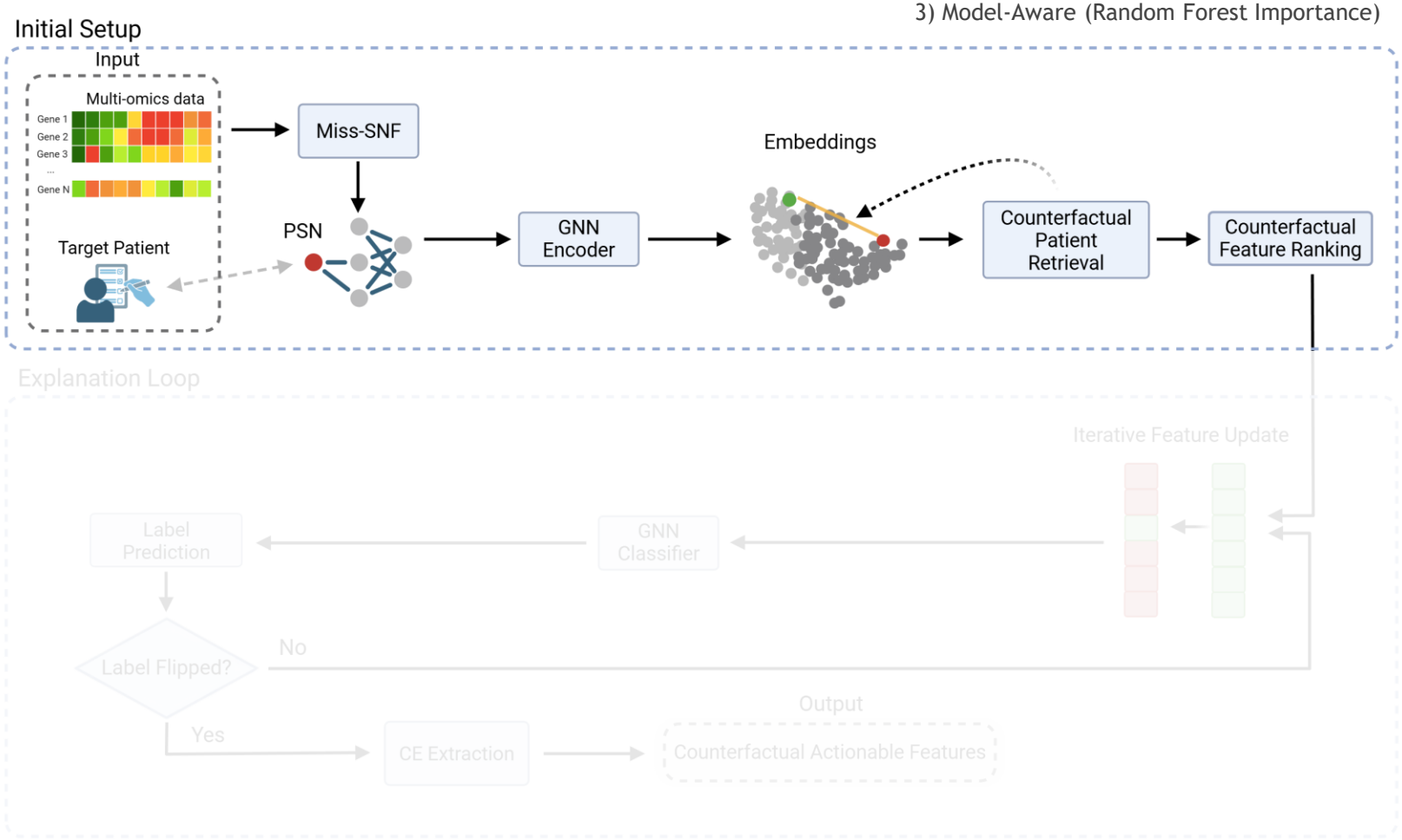
CE-GNN: Initial Setup

Contrastive Loss: pulls same-outcome patients together, pushes different outcomes apart

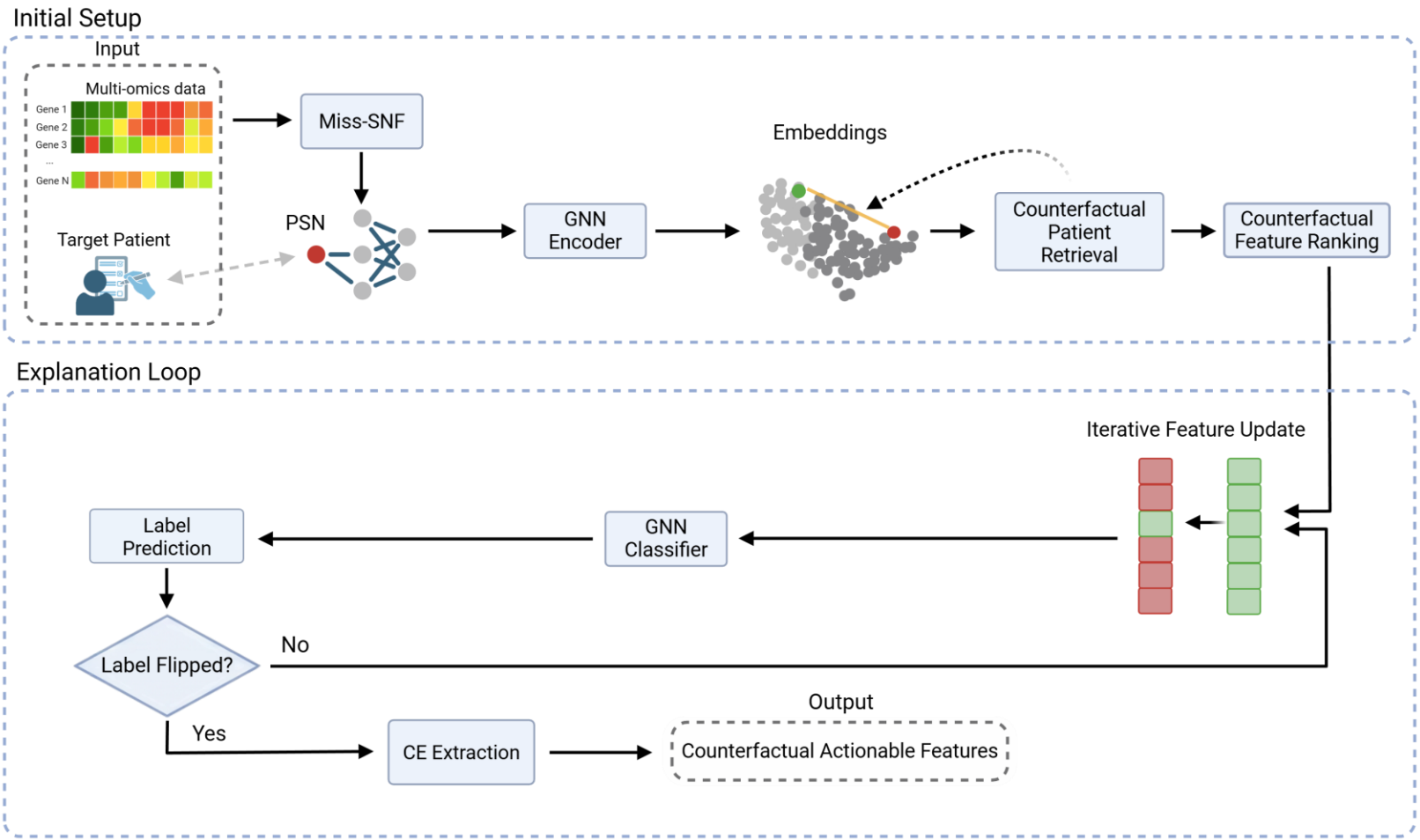


CE-GNN: Patient Retrieval & Feature Ranking

Ranking strategies: 1) Absolute Difference ($|f^i - f^{ci}|$),
2) Statistical Significance (ANOVA F-value),
3) Model-Aware (Random Forest Importance)



CE-GNN: Explanation Loop



Experimental Setup

- **Dataset:**
  - **Synthetic** Networked Data (with known ground truth)
  - **ROSMAP** (Alzheimer's Disease (AD) dataset, curated by MOGONET [Wang '21])
- **Proposed framework:**
  - **CE-GNN** with 3 ranking strategies: Absolute Difference (Abs), Statistical (St), Random Forest (RF)
- **Baselines:**
  - Counterfactual tabular-based: **DiCE** [Mothilal '20], **SingleCF** [Wachter '18]
  - Factual graph-based: **GNNExplainer** [Ying '19]
- **Metrics:**
  - **Ground truth** recovery: Precision, Recall, Macro-F1, Accuracy
  - **Quality** of generated **counterfactuals**: Validity, Fidelity, Sparsity

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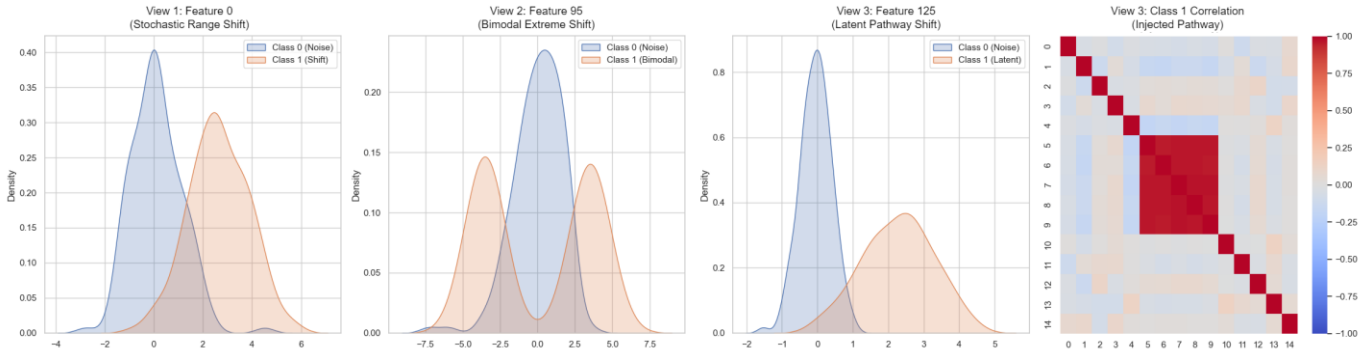
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Validation on Synthetic Data

Method	Precision	Recall	macro-F1	Accuracy	Validity	Fidelity	Sparsity
GNNExplainer	57.31	35.89	44.14	**89.48**	-	-	**92.30**
SingleCF	10.54	**52.33**	17.55	51.44	**100.0**	**96.67**	50.98
DiCE	10.42	51.22	17.32	51.71	**100.0**	**96.67**	51.47
CE-GNN (Abs.)	11.94	37.11	18.07	73.76	98.33	95.00	76.17
CE-GNN (St.)	46.36	34.78	39.74	84.60	**100.0**	**96.67**	87.64
CE-GNN (RF)	**60.59**	38.22	**46.87**	86.73	**100.0**	**96.67**	89.09



Real-World Results: ROSMAP

Method	Validity	Fidelity	Sparsity
GNNExplainer	-	-	80.17
SingleCF	**100.0**	**70.75**	49.69
DiCE	**100.0**	**70.75**	51.24
CE-GNN (Abs.)	91.51	62.26	**85.75**
CE-GNN (St.)	85.85	56.60	75.69
CE-GNN (RF)	90.57	61.32	77.95

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Biological validation: most of the identified features align with known AD biomarkers in literature

Limitations & Future Work

Current **limitations**:

- Sub-optimal Validity on ROSMAP (< 100%)
- **Static topology** bottleneck (only node features are perturbed)

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Future ongoing **directions**:

- **Dynamic** CE-GNN
- Broader experimental benchmarking

Conclusions: CE-GNN

- **First** CE framework tailored for Patient Similarity Networks
- It is able to identify **actionable features** for **specific therapeutics** in a clinical setting
- Ongoing work on a **dynamic version** of CE-GNN perturbing both node features and network topology to achieve full validity

Thank you for your attention!

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Preprint